

α/β -Hydrolase domain-containing 6 (ABHD6) accelerates the desensitization and deactivation of TARP γ -2-containing AMPA receptors

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Abstract

AMPA receptors (AMPARs) mediate most of the fast excitatory synaptic transmission in mammalian brain. Their efficacy in responding to presynaptic glutamate release depends on their kinetics, which are determined by AMPARs and their auxiliary subunit composition. α/β -Hydrolase domain-containing 6 (ABHD6) is an AMPAR auxiliary subunit that has been shown to negatively regulate the surface delivery of AMPARs and AMPAR-mediated currents. Overexpression of ABHD6 decreased the rising slope and increased the decay τ of mEPSCs. However, whether ABHD6 is involved in regulating AMPAR kinetics remains unclear. Here, we found that ABHD6 per se had no effects on the gating kinetics of GluA1 and GluA2(Q) containing homomeric receptors. However, in the presence of the auxiliary subunit TARP γ -2, ABHD6 accelerated the deactivation and desensitization of either GluA1 and GluA2(Q) containing homomeric receptors independent of their splicing isoforms (flip and flop) and the editing isoforms of GluA2 (R or G at position 764), except the deactivation of GluA2(Q)i-G isoform. Besides, the recovery from desensitization of GluA1 with flip splicing isoform was slowed by the co-expression of ABHD6 in the presence of TARP γ -2. Furthermore, the ABHD6 accelerated the deactivation and desensitization of GluA1i/GluA2(R)i-G heteromeric receptors in the presence of TARP γ -2. Therefore, these results demonstrate that ABHD6 regulates AMPAR gating kinetics in a TARP γ -2-dependent manner.

Significance statement

The efficacy of AMPARs in responding to presynaptic glutamate release depends on their kinetics, including deactivation, desensitization, and recovery from desensitization, which are determined by AMPARs and their auxiliary subunit composition. Using ultra-fast

application of glutamate and outside-out patch recordings, we found that, in the presence of the auxiliary subunit TARP γ -2, ABHD6 accelerated the deactivation and desensitization of GluA1i/GluA2(R)i-G heteromeric receptors and GluA1 and GluA2(Q) containing homomeric receptors independent of their splicing isoforms (flip and flop) and the editing isoforms of GluA2 (R or G at position 764), except the deactivation of GluA2(Q)i-G isoform. These results demonstrate that ABHD6 regulates AMPAR gating kinetics in a TARP γ -2-dependent manner.

eLife assessment

This **fundamental** work demonstrates that ABHD6 regulates AMPAR gating kinetics in a TARP γ -2-dependent manner. The evidence in this study is **compelling**. This study will be of interest to readers in the field of synaptic transmission.

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Introduction

The α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors (AMPA), tetramers consisting of GluA1-4 subunits, are glutamate-gated ion channels that mediate fast excitatory synaptic transmission in the central nervous system (Bredt and Nicoll, 2003; Diering and Huganir, 2018). The transmission strength depends on the number and kinetic characteristics of AMPARs at the postsynaptic membrane (Huganir and Nicoll, 2013). AMPARs exhibit fast kinetics, with activation, deactivation, and desensitization in milliseconds, which enable the rapid depolarization of the postsynaptic membrane, ensuring a high speed and fidelity of signaling in the nervous system to accommodate the speed of information processing in the brain (Erreger et al., 2004; Mayer, 2005; Luscher and Malenka, 2012; Greger et al., 2017). AMPAR kinetics not only affects synaptic transmission and plasticity, but also neurotoxicity through Ca^{2+} permeability (Huganir and Nicoll, 2013; Diering and Huganir, 2018; Yang et al., 2021). Abnormalities in AMPAR kinetics disrupt synaptic signaling and neuronal homeostasis, which are associated with a range of neurological disorders, such as Alzheimer's disease, epilepsy, amyotrophic lateral sclerosis, and stroke (Qneibi et al., 2019; Yang et al., 2021).

In the brain, auxiliary subunits are found to assemble with AMPARs with different stoichiometries, and modulate AMPAR kinetics (Kato et al., 2008; Brockie and Maricq, 2010; Cheng et al., 2012; Schwenk et al., 2012; Shanks NF, 2012; Yang et al., 2014; Matt et al., 2018). TARP γ -2 was the first auxiliary subunit to be discovered. This subunit positively regulates AMPAR functions, such as promoting affinity, increasing surface expression, slowing deactivation and desensitization, and accelerating recovery from desensitization (Gill et al., 2011; Chen et al., 2017). Besides, other auxiliary proteins have also been reported to modulate AMPAR kinetics. CNH-2 slows the deactivation and desensitization of GluA1i/A2(R) in *Xenopus* oocytes but has no effect on its recovery from desensitization, while it slows the deactivation and desensitization of GluA1i and accelerates its recovery from desensitization in HEK 293T cells. GSG1L slows AMPAR deactivation, desensitization, and recovery from desensitization in *Xenopus* oocytes and HEK cells (Kato et al., 2010; Khodosevich et al., 2014; Jacobi and von Engelhardt, 2017). This diversity results from coassembly of the known AMPAR subunits, pore-forming GluA and three types of auxiliary proteins, with 21 additional constituents, mostly secreted proteins or transmembrane proteins of different classes. Their integration at distinct abundance and stability establishes the heteromultimeric architecture of native AMPAR complexes: a defined core with a variable periphery resulting in an apparent molecular mass between 0.6 and 1 MDa (Schwenk et al., 2012; Khodosevich et al., 2014). Studies have found

that TARP γ -8 and CNIH2 could bound to the same AMPARs complex, and compared to existence of TARP γ -8 alone with GluA1, the time course of AMPAR deactivation and decreased desensitization decreased with CNIH-2 affiliated. And the binding sites for these two families (TARPs and CNIHs) of transmembrane proteins are distinct (Shi et al., 2010 [DOI](#)). Besides, using cryo-electron microscopy (cryo-EM), studies showed that TARP γ -8 and CNIH2 can assemble in the same AMPAR complex (Yu et al., 2021 [DOI](#)).

ABHD6 is a member of the α/β -hydrolase family. It was previously reported to be a serine hydrolase that could control 2-arachidonoylglycerol (2-AG) levels at cannabinoid receptors, plays a critical role in 2-AG signaling in endocannabinoid signaling system (Chevaleyre et al., 2006 [DOI](#); Marrs et al., 2010 [DOI](#)). High-resolution proteomic analysis has shown that ABHD6 is associated with the natural AMPAR core subunit (Schwenk et al., 2012 [DOI](#)). Functional studies revealed that overexpression of ABHD6 drastically reduced AMPAR-mediated currents by selectively inhibits the surface expression levels of AMPARs without altering total expression levels of AMPARs. The effects of ABHD6 on AMPARs was observed in both neurons and heterologous cells (Wei et al., 2016 [DOI](#); Wei et al., 2017 [DOI](#)). Wei et al. also tested whether the observed effects was due to ABHD6's hydrolase activity. Using the specific inhibitor of ABHD6 lipase activity, WWL70, or the lipase activity deficit mutation (S148A) of ABHD6, they found that the effects was independent of ABHD6's hydrolase activity. Using two-dimensional gel separation of the ER and membrane fractions, Schwenk and colleagues demonstrated the assembly of AMPARs in ER and their delivery to plasma membrane (Schwenk et al., 2019 [DOI](#)). They showed ABHD6 trapped GluAs in the monomeric GluA-ABHD6 complex, and thus eliminated tetramer formation and surface delivery of AMPARs. In addition to this, overexpression of ABHD6 significantly increased the decay τ of mEPSCs in cultured hippocampal neurons, which indicated that ABHD6 may affect the kinetics of AMPARs (Wei et al., 2016 [DOI](#); Wei et al., 2017 [DOI](#)). However, whether and how ABHD6 regulates AMPAR kinetics remain unknown. In this study, we investigated whether ABHD6 plays a regulatory role in the kinetic properties of functional AMPARs and whether this role is subject to AMPAR RNA editing and splice variants differences. Following previous studies (Priel et al., 2005 [DOI](#); Milstein et al., 2007 [DOI](#); Kato et al., 2008 [DOI](#); Gill et al., 2011 [DOI](#); Khodosevich et al., 2014 [DOI](#)), we transfected different GluAs, GluAs + ABHD6, GluAs + TARP γ -2, GluAs + TARP γ -2 + ABHD6 in HEK 293T cells, and systematically studied their channel properties. Our data showed that ABHD6 did not affect the kinetic properties of AMPARs. However, in the presence of TARP γ -2, ABHD6 accelerated the deactivation and desensitization of TARP γ -2 containing GluA1 and GluA2(Q) homomeric receptors independent of their splicing isoforms and editing isoforms, except the deactivation of GluA2(Q)i-G isoform, and the deactivation and desensitization of GluA1i/GluA2(R)i-G heteromeric receptors. Our findings revealed a new function of ABHD6 in regulating the channel properties TARP γ -2-containing AMPA receptors.

Results

ABHD6 reduced the glutamate-induced currents mediated by AMPARs in the presence and absence of TARP γ -2

Previous studies have shown that ABHD6 can significantly decrease currents in HEK 293T cells transfected with GluA1i, GluA2(R)i, GluA3i, GluA1i+GluA2(R)i, GluA2(R)i+GluA3i, or GluA2(R)i+GluA3i in the presence or absence of TARP γ -2 (Wei et al., 2016 [DOI](#); Wei et al., 2017 [DOI](#); Schwenk et al., 2019 [DOI](#)). However, whether ABHD6 inhibits other RNA editing and splice variants of AMPARs remains unclear. To answer this question, we constructed a total of 14 isoforms of GluA1, GluA2, and GluA3, specifically GluA1i, GluA1o, GluA2(Q)i-R, GluA2(Q)o-R, GluA2(Q)i-G, GluA2(Q)o-G, GluA2(R)i-R, GluA2(R)o-R, GluA2(R)i-G, GluA2(R)o-G, GluA3i-R, GluA3o-R, GluA3i-G, and GluA3o-G (illustrated in **Fig. 1 A** [DOI](#) and **Fig. EV1** [DOI](#)). We recorded glutamate-induced currents in HEK 293T cells transfected with these variants together with ABHD6 or control plasmid. Our results showed that ABHD6 overexpression decreased the peak amplitudes of the currents

mediated by GluA1i to ~34%, GluA1o to ~48%, GluA2(Q)i-R to ~31%, GluA2(Q)o-R to ~32%, GluA2(Q)i-G to ~45%, and GluA2(Q)o-G to ~21% (**Fig. 1 B-C**, **Table. EV1.1**, **EV1.2**). We also recorded the currents of these variants in the presence of TARP γ -2. TARP γ -2 increased GluA1- and GluA2-induced currents, but its effect on the increase in GluA3-induced currents was not significant, which is generally consistent with previous studies reporting a small effect of TARP γ -2 on GluA3 currents (Pei et al., 2007; Coleman et al., 2016). Moreover, ABHD6 overexpression decreased the peak amplitudes of currents mediated by GluA1i to ~14%, GluA1o to ~37%, GluA2(Q)i-R to ~9%, GluA2(Q)o-R to ~11%, GluA2(Q)i-G to ~6%, GluA2(Q)o-G to ~7%, GluA2(R)i-R to ~10%, GluA2(R)o-R to ~44%, GluA2(R)i-G to ~11%, GluA2(R)o-G to ~21%, GluA3i-R to ~5%, GluA3o-R to ~38%, GluA3i-G to ~21%, and GluA3o-G to ~24%, in the presence of TARP γ -2 (**Fig. 1 B-D**, **Table. EV1.1**, **EV1.2**). Taken together, these results showed that ABHD6 significantly reduced the glutamate-induced currents mediated by AMPARs in the pre- or ab-sence of TARP γ -2 independently of the subunit type, flip/flop splice variants, and Q/R or R/G RNA editing.

ABHD6 accelerated the deactivation of homomeric AMPAR-TARP γ -2 complexes

To investigate whether ABHD6 affected AMPAR deactivation in the pre- or ab-sence of TARP γ -2, we transfected HEK 293T cells with GluAs (GluA1i, GluA1o, GluA2(Q)i-R, GluA2(Q)o-R, GluA2(Q)i-G, GluA2(Q)o-G, GluA1i/GluA2(R)i), GluAs + ABHD6, GluAs + TARP γ -2, and GluAs + ABHD6 + TARP γ -2. 1-ms application of 10 mM glutamate could induce the receptor deactivation, a rapidly activated inward current followed by rapid decay to baseline levels, under outside-out patch recordings (-60 mV). Concerning about impact of GluA expression levels on the membrane on the current kinetics, we conducted Pearson correlation analyses between the amplitude and kinetics of the currents from various glutamate receptor assemblies we measured, results showed that there is no correlation between peak amplitude and kinetics of AMPARs (**Fig. EV6**). Consistent with previous observations, our data showed that TARP γ -2 slowed AMPAR deactivation without selective differences in subunits, splice variants, or RNA editing (**Fig. 2**). Moreover, ABHD6 overexpression had a negligible effect on the deactivation of AMPARs, including GluA1i, GluA1o, GluA2(Q)i-R, GluA2(Q)o-R, GluA2(Q)i-G, and GluA2(Q)o-G (**Fig. 2 A-F**, **Fig. EV2 A-F**, **Table. EV2.1**, **EV2.2**). However, ABHD6 overexpression reduced the deactivation time constant ($\tau_{w, deac}$) of AMPAR in the presence of TARP γ -2, except for GluA2(Q)i-G (**Fig. 2 A-F**, **Fig. EV2 A-F**, **Table. EV2.1**, **EV2.2**). Specifically, ABHD6 significantly decreased the $\tau_{w, deac}$ of GluA1i to ~48%, of GluA1o to ~65%, of GluA2(Q)i-R to ~48%, of GluA2(Q)o-R to ~60%, and of GluA2(Q)o-G to ~60% in the presence of TARP γ -2 (**Fig. 2 A-F**, **Fig. EV2 A-F**, **Table. EV2.1**, **EV2.2**).

Next, we wonder whether the observed phenotype was due to the effects of ABHD6 on TARPed AMPARs, or due to the changes of the ratio of TARPed and unTARPed receptors under our experimental condition. First of all, to avoid that the observed phenotype was due to the changes of the ratio of TARPed to unTARPed AMPARs following ABHD6 overexpression, we constructed chimera plasmids that fused GluA2(Q)i-R or GluA1i and TARP γ -2 together. We recorded the deactivation using outside-out patch recordings (-60 mV), and examined whether ABHD6 could affect its kinetics. We found that overexpression of ABHD6 decreased the peak amplitudes of the currents of GluA2(Q)i-R-TARP γ -2 to ~23%, and GluA1i-TARP γ -2 to ~25%, and accelerated $\tau_{w, deac}$ mediated by GluA2(Q)i-R-TARP γ -2 to ~65% or GluA1i-TARP γ -2 to ~55% (**Fig. 2G**, **2I**, **Fig. EV2G**, **2I**, **Table. EV2.1**, **EV2.2**). These results are consistent with our previous findings using two plasmids to express AMPARs and TARP γ -2. Secondly, we isolated TARP γ -2-containing AMPA receptors using the method of Carbone et al., Nature Communications, 2016, in which they added 50 μ M spermine to the internal solution and recorded at +50 mV. We transfected chimera plasmids GluA2(Q)i-R-TARP γ -2 or GluA1i-TARP γ -2 together with ABHD6 or control plasmid, and recorded the deactivation of AMPA receptors at +50 mV in the presence of 50 μ M spermine. Our results showed that ABHD6 could still decreased the peak amplitudes of currents and accelerated

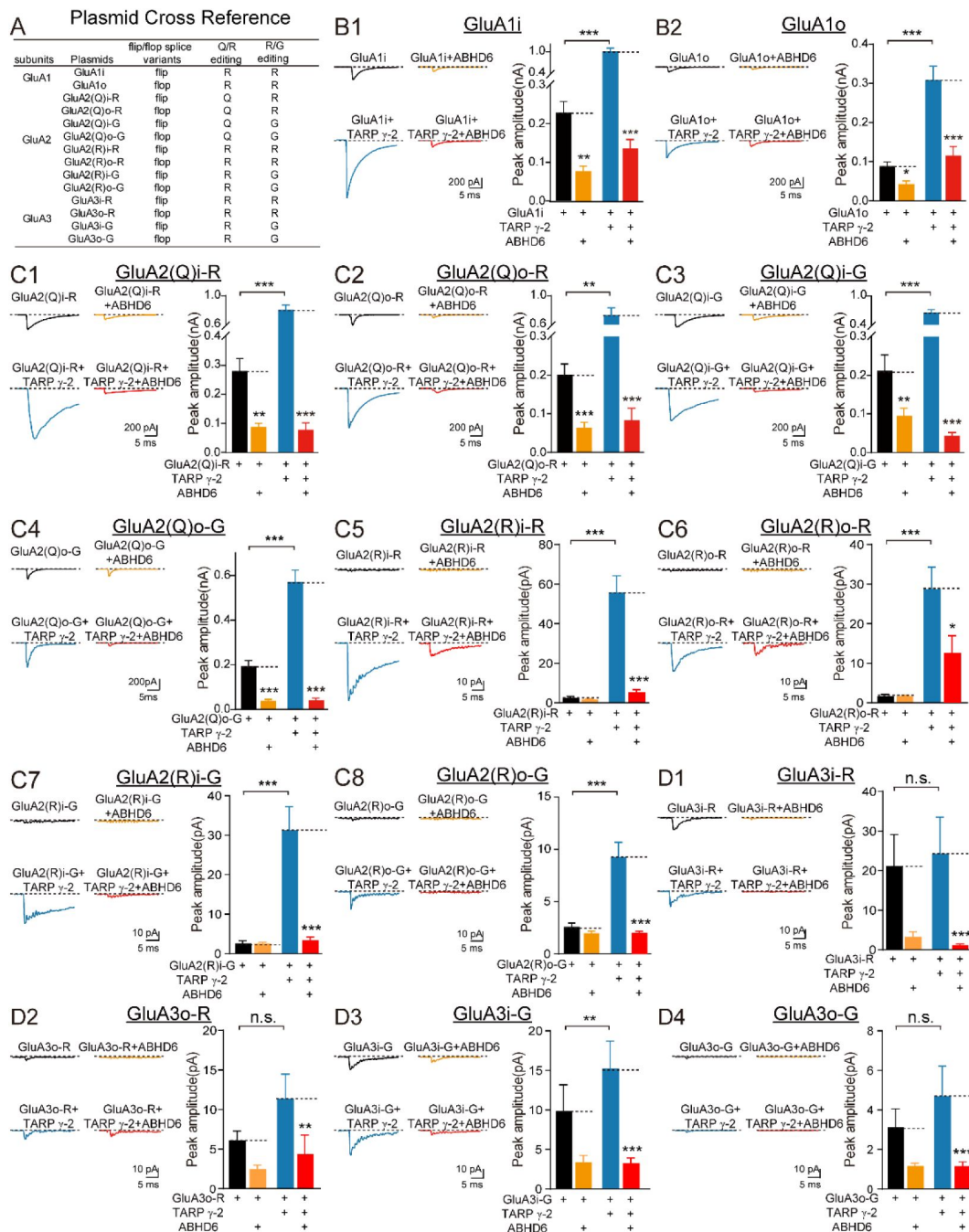


Figure 1.

the Effect of overexpression of ABHD6 on the reduction of peak current in AMPARs.

(A) Plasmid abbreviations and variant combinations of AMPAR.

(B-D) Representative traces (left) and summary graphs of the peak amplitudes (right) of 10 mM glutamate-induced currents in HEK 293T cells transfected with GluA1-3 (black), GluA1-3 + ABHD6 (orange), GluA1-3 + TARP γ -2 (blue), GluA1-3 + TARP γ -2 + ABHD6 (red).

The statistical method was one-way ANOVA followed by a two-way comparison (* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. **Table. EV1.2**).

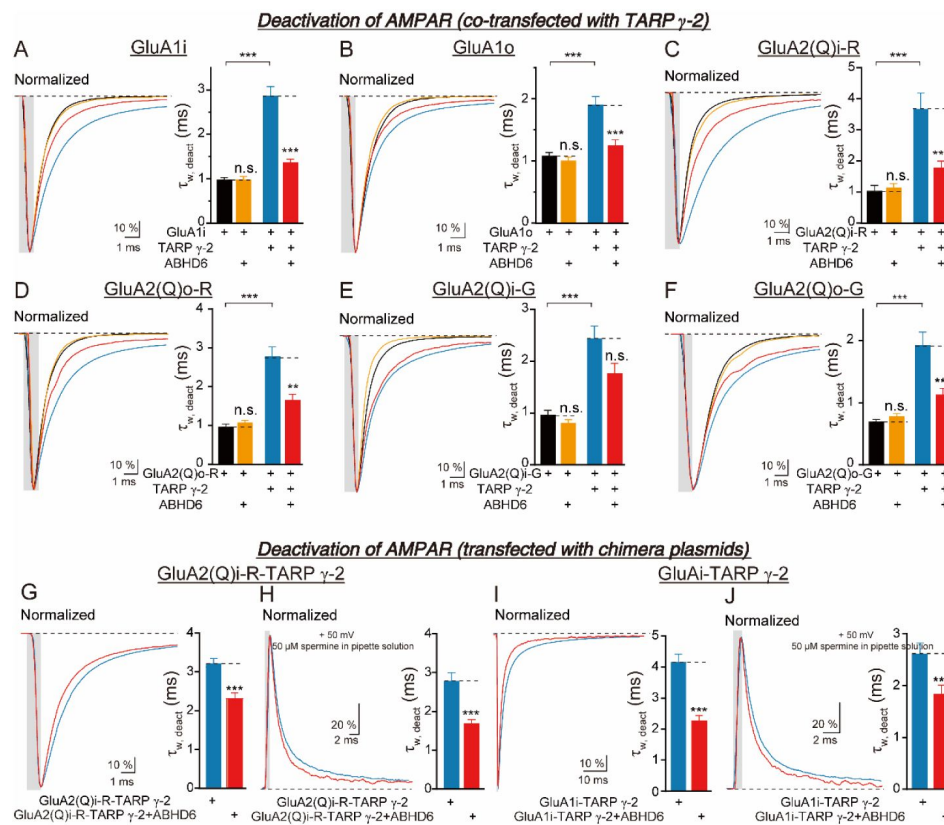


Figure 2.

Overexpression of ABHD6 accelerated the deactivation of AMPARs-TARP γ -2 complexes in HEK 293T cells.

The normalized traces, and the summary bar graphs of the $\tau_{w,deact}$ of Glutamate (10 mM Glu, 1 ms) induced currents in the outside-out patch from HEK 293T cells transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G) GluA2(Q)j-i-R-TARP γ -2 tandem. (I) GluA1i-TARP γ -2 tandem. γ -2-containing GluA receptors could be isolated when 50 μ M spermine in the internal solution and recorded at +50 mV, the average traces and the normalized traces (right), and the summary bar graphs of the $\tau_{w,deact}$ of Glutamate (10 mM Glu, 1 ms) induced currents in the outside-out patch recorded at +50 mV from HEK 293T cells transfected with GluA-TARP γ -2 tandem (blue), GluA-TARP γ -2 tandem + ABHD6 (red). (H) GluA2(Q)j-i-R-TARP γ -2. (J) GluA1i-TARP γ -2. The statistical method was one-way ANOVA followed by a two-way comparison (* P < 0.05; ** P < 0.01; *** P < 0.001. **Table. EV2.2** [Table. EV2.2](#)).

the $\tau_{w,deact}$ mediated by GluA2(Q)i-R-TARP γ -2 or GluA1i-TARP γ -2 (Fig. 2H [↗](#), 2J [↗](#), Fig. EV2H [↗](#), 2J [↗](#), Table. EV2.1 [↗](#), EV2.2 [↗](#)). Taken together, our results clearly show that ABHD6 could affect the gating kinetics of TARPed AMPARs.

Collectively, these results showed that ABHD6 accelerated the deactivation of TARP γ -2-containing AMPARs, except for the GluA2(Q)i-G variants.

ABHD6 accelerated the desensitization of homomeric AMPAR-TARP γ -2 complexes

To investigate whether ABHD6 affected AMPAR desensitization, we recorded the desensitization induced by a 100-ms application of 10 mM glutamate using outside-out patch recordings (-60 mV). The desensitization curve showed a slow decay pattern than deactivation due to the long-term action of glutamate. ABHD6 overexpression had an insignificant effect on the desensitization of AMPARs, including GluA1i, GluA1o, GluA2(Q)i-R, GluA2(Q)o-R, GluA2(Q)i-G, and GluA2(Q)o-G (Fig. 3 A-F [↗](#), Fig. EV3 A-F [↗](#), Table. EV3.1 [↗](#), EV3.2 [↗](#)). We also studied the effects of ABHD6 on the desensitization in the presence of TARP γ -2. Consistent with previous observations, our data showed that TARP γ -2 slowed AMPAR desensitization without selective differences in subunits, splice variants, or RNA editing (Priel et al., 2005 [↗](#); Coombs et al., 2012 [↗](#); Devi et al., 2020 [↗](#)). ABHD6 overexpression reduced the desensitization time constant ($\tau_{w,des}$) of AMPARs in the presence of TARP γ -2. Specifically, it significantly decreased the $\tau_{w,des}$ of GluA1i to ~62%, of GluA1o to ~63%, of GluA2(Q)i-R to ~59%, of GluA2(Q)o-R to ~65%, of GluA2(Q)i-G to ~60%, and of GluA2(Q)o-G to ~71% (Fig. 3 A-F [↗](#), Fig. EV3 A-F [↗](#), Table. EV3.1 [↗](#), EV3.2 [↗](#)) in the presence of TARP γ -2. To further confirm that the observed phenotype of ABHD6 on AMPAR desensitization was due to the effect of ABHD6 on TARPed AMPARs, we also tried the chimera plasmids that fused GluA2(Q)i-R or GluAi and TARP γ -2 together as we did in deactivation studies. We found that overexpression of ABHD6 accelerated $\tau_{w,des}$ mediated by GluA2(Q)i-R-TARP γ -2 or GluA1i-TARP γ -2 (Figure 3G [↗](#), 3I [↗](#), Figure EV3G [↗](#), 3I [↗](#), Table. EV3.1 [↗](#), EV3.2 [↗](#)). Besides, we added 50 μ M spermine to the internal solution and recorded at +50 mV to isolated TARPed receptors, and found that ABHD6 could still accelerated the $\tau_{w,des}$ mediated by GluA2(Q)i-R-TARP γ -2 or GluA1i-TARP γ -2 (Figure 3H [↗](#), 3J [↗](#), Figure EV3H [↗](#), 3J [↗](#), Table. EV3.1 [↗](#), EV3.2 [↗](#)).

Collectively, these results showed that ABHD6 accelerated the desensitization of TARP γ -2-containing AMPARs and that the effect did not depend on differences in GluA1/2(Q) subunits, flip/flop splice variants, or R/G editing.

ABHD6 slowed the recovery of homomeric GluA1i-TARP γ -2 complexes from desensitization

Recovery from desensitization is another important parameter of AMPAR channel kinetics. To investigate whether ABHD6 affected AMPAR recovery from desensitization, we recorded the recovery curve in the presence or absence of TARP γ -2. Using outside-out patch recordings (-60 mV), we performed 100-ms paired-pulse stimulation at different intervals (stimulation was performed at intervals of 1-601 ms) to plot the recovery ratio curve. Consistent with previous studies, our data showed that TARP γ -2 accelerated GluA1 recovery from desensitization (Fig. 4 A-B [↗](#), Fig. EV4 A-B [↗](#), Table. EV4.1 [↗](#), EV4.2 [↗](#)). Moreover, it accelerated GluA2(Q)i but not GluA2(Q)o recovery (Fig. 4 C-F [↗](#), Fig. EV4 C-F [↗](#), Table. EV4.1 [↗](#), EV4.2 [↗](#)). We also found that ABHD6 overexpression had little effect on the recovery of AMPARs in the presence or absence of TARP γ -2, except for GluA1i-TARP γ -2, whose recovery from desensitization time constant (τ_{rec}) increased to ~157% (Fig. 4 A [↗](#), Fig. EV4 A [↗](#), Table. EV4.1 [↗](#), EV4.2 [↗](#)). We confirmed this results using chimera GluA2(Q)i-R-TARP γ -2 and GluA1i-TARP γ -2 plasmid, and found that ABHD6 overexpression increased the τ_{rec} of GluA1i-TARP γ -2 to ~135%, without changing the τ_{rec} of GluA2(Q)i-R-TARP γ -2 (Fig. 4 G-H [↗](#), Fig. EV4 G-H [↗](#), Table. EV4.1 [↗](#), EV4.2 [↗](#)). These results

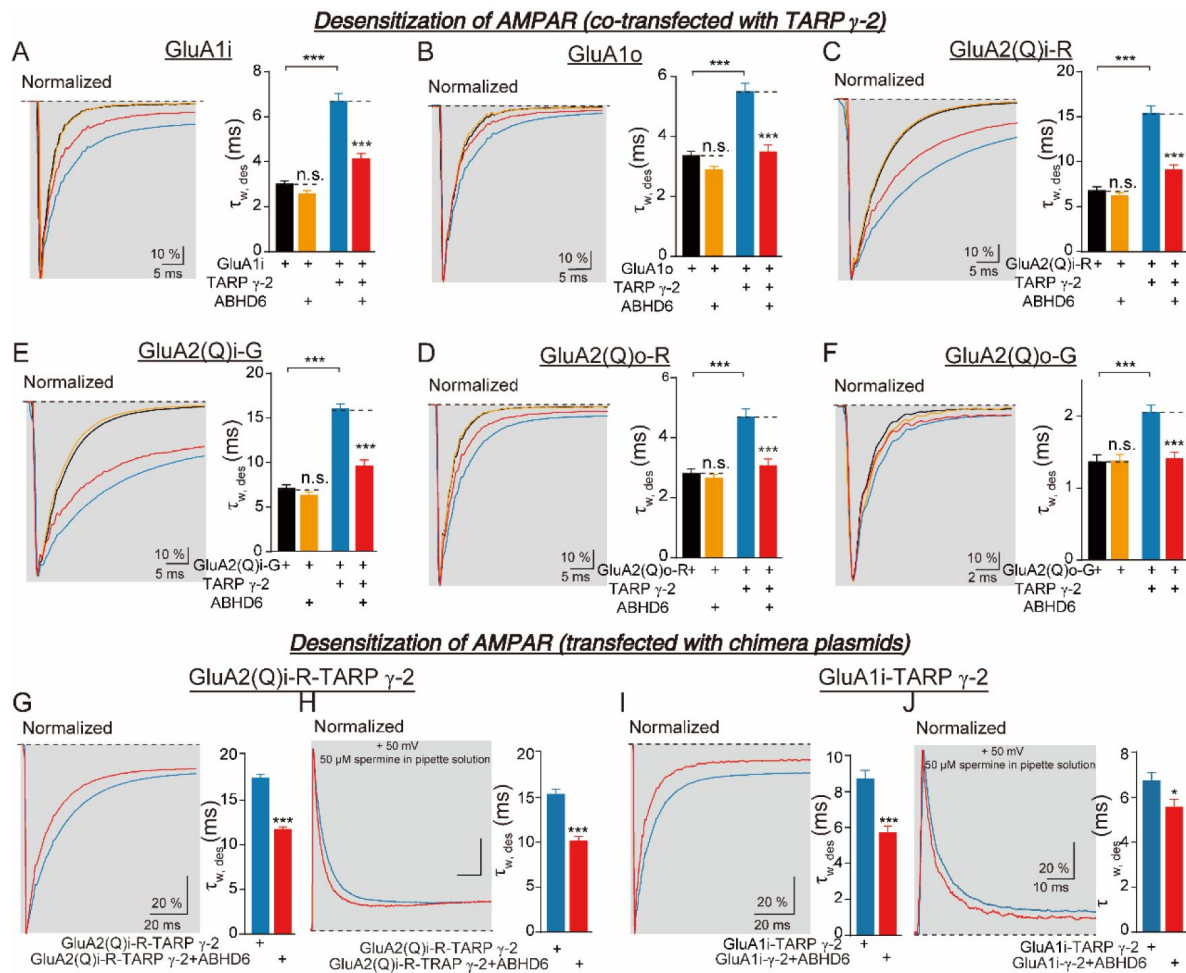


Figure 3.

Overexpression of ABHD6 accelerated the desensitization of AMPARs-TARP γ -2 complexes in HEK 293T cells.

The normalized traces, and the summary bar graphs of the $\tau_{w,des}$ of Glutamate (10 mM Glu, 100 ms) induced currents in the outside-out patch from HEK 293T cells transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G) GluA2(Q)i-R-TARP γ -2 tandem. (I) GluA1i-TARP γ -2 tandem. TARP γ -2-containing GluA receptors could be isolated when 50 μ M spermine in the internal solution and recorded at +50 mV, the average traces and the normalized traces (right), and the summary bar graphs of the $\tau_{w,des}$ and peak amplitude of Glutamate (10 mM Glu, 100 ms) induced currents in the outside-out patch recorded at +50 mV from HEK 293T cells transfected with GluA-TARP γ -2 tandem (blue), GluA-TARP γ -2 tandem + ABHD6 (red). (H) GluA2(Q)i-R-TARP γ -2. (J) GluA1i-TARP γ -2. The statistical method was one-way ANOVA followed by a two-way comparison (* P < 0.05; ** P < 0.01; *** P < 0.001. [Table EV3.2](#)).

showed that the GluA1i-TARP γ -2 complex played a specific role in the regulation of AMPAR recovery from desensitization by ABHD6. The recovery of flip splice variants was more susceptible to regulation by other auxiliary subunits.

ABHD6 negatively regulates the kinetics of heteromeric GluA1i/GluA2(R)i-G-TARP γ -2 complexes

To investigate whether affected kinetic properties of native receptor subtypes in near-physiological states, we performed additional experiments using heteromeric receptors. Previous studies have shown that diheteromeric GluA1/GluA2 and GluA2/GluA3 receptors are the most common sets in the mammalian brain, especially in the mouse hippocampus where the diheteromeric GluA1/GluA2 occupies more than 50% of the calcium-impermeable AMPA receptor complexes (Wenthold et al., 1996 [\[1\]](#); Lu et al., 2009 [\[2\]](#); Zhao et al., 2019 [\[3\]](#); Yu et al., 2021 [\[4\]](#)). Following previous studies (Cho et al., 2007 [\[5\]](#); Schwenk et al., 2009 [\[6\]](#); Schwenk et al., 2012 [\[7\]](#); Herring et al., 2013 [\[8\]](#); Klaassen et al., 2016 [\[9\]](#)), We co-transfected GluA1i/GluA2(R)i-G, GluA1i/GluA2(R)i-G+ABHD6, GluA1i/GluA2(R)i-G+ TARP γ -2, GluA1i/GluA2(R)i-G+ TARP γ -2+ABHD6 in HEK 293T cells (GluA1i: GluA2(R)i-G: TARP γ -2 : ABHD6=0.4 μ g : 0.4 μ g : 1.2 μ g : 30 ng). And we recorded the deactivation and desensitization and recovery as previously described using outside-out patch recordings (-60 mV). Consistent with previous studies (Schwenk et al., 2009 [\[6\]](#); Coombs et al., 2012 [\[10\]](#)), our data showed that TARP γ -2 increased the peak amplitude of currents mediated by GluA1i/GluA2(R)i-G receptors, slowed the deactivation and desensitization of GluA1i/GluA2(R)i-G receptors and accelerated its recovery (Fig. 5 [\[11\]](#), Fig. EV5 [\[12\]](#), Table. EV5.1 [\[13\]](#), EV5.2 [\[14\]](#)). Our results showed that ABHD6 only accelerated the deactivation and desensitization of TARP γ -2-containing GluA1i/GluA2(R)i-G receptors, $\tau_{w, deact}$ and $\tau_{w, des}$ of GluA1i/GluA2(R)i-G-TARP γ -2 to ~62% and ~55% respectively (Fig. 5A [\[15\]](#), 5B [\[16\]](#), Table. EV5.1 [\[13\]](#), EV5.2 [\[14\]](#)). However, ABHD6 overexpression decreased the currents and slowed its recovery in the presence and absence of TARP γ -2. Specifically, ABHD6 significantly decreased the peak amplitudes of currents mediated by GluA1i/GluA2(R)i-G receptor to ~62%, of GluA1i/GluA2(R)i-G-TARP γ -2 to ~17%. ABHD6 increased the τ_{rec} of GluA1i/GluA2(R)i-G and GluA1i/GluA2(R)i-G-TARP γ -2 to ~203% and ~198% (Fig. 5C [\[17\]](#), Fig. EV5 [\[12\]](#), Table. EV5.1 [\[13\]](#), EV5.2 [\[14\]](#)).

Discussion

In this study, we systematically studied the effects of ABHD6 on the amplitudes and kinetics of AMPARs with different subunit types, GluA1(flip/flop), GluA2(Q) (flip/flop, R/G) and diheteromeric GluA1i/GluA2(R)i-G receptors, in the presence or absence of TARP γ -2 in HEK 293T cells. Our results showed that ABHD6 inhibited glutamate-induced currents in all different AMPARs. ABHD6 regulated the gating kinetics of AMPARs in the presence of TARP γ -2.

Previous studies have reported that ABHD6 can reduce glutamate-induced currents and the surface expression of GluAs independently of the subunit composition (Wei et al., 2016 [\[18\]](#); Wei et al., 2017 [\[19\]](#)). They found that ABHD6 overexpression can reduce AMPAR-currents and surface expression levels of AMPARs in HEK 293T cells transfected with GluA1i, GluA2(R)i, GluA3i, GluA1i+GluA2(R)i, GluA2(R)i+GluA3i, or GluA2(R)i+GluA3i in the presence or absence of TARP γ -2. In addition to flip splice of AMPAR subunits, mammalian brains also have the flop splicing isoform. The flop splice variants are expressed in a pattern that differs from but partially overlaps with flip splice variants. The CA1 and CA3 regions of the hippocampus predominantly express flip splice variants, whereas DG highly expresses flop splice variants (Sommer et al., 1990 [\[20\]](#)). Flip splice variants remain unchanged in the developing brain since birth, whereas flop splice variants are expressed at low levels in the first eight days of life and reach adult levels by day 14, suggesting that flop splice variants may be involved in the formation of mature receptors (Monyer et al., 1991 [\[21\]](#)). Q/R editing enables the conversion of neutral to positively charged residues in the

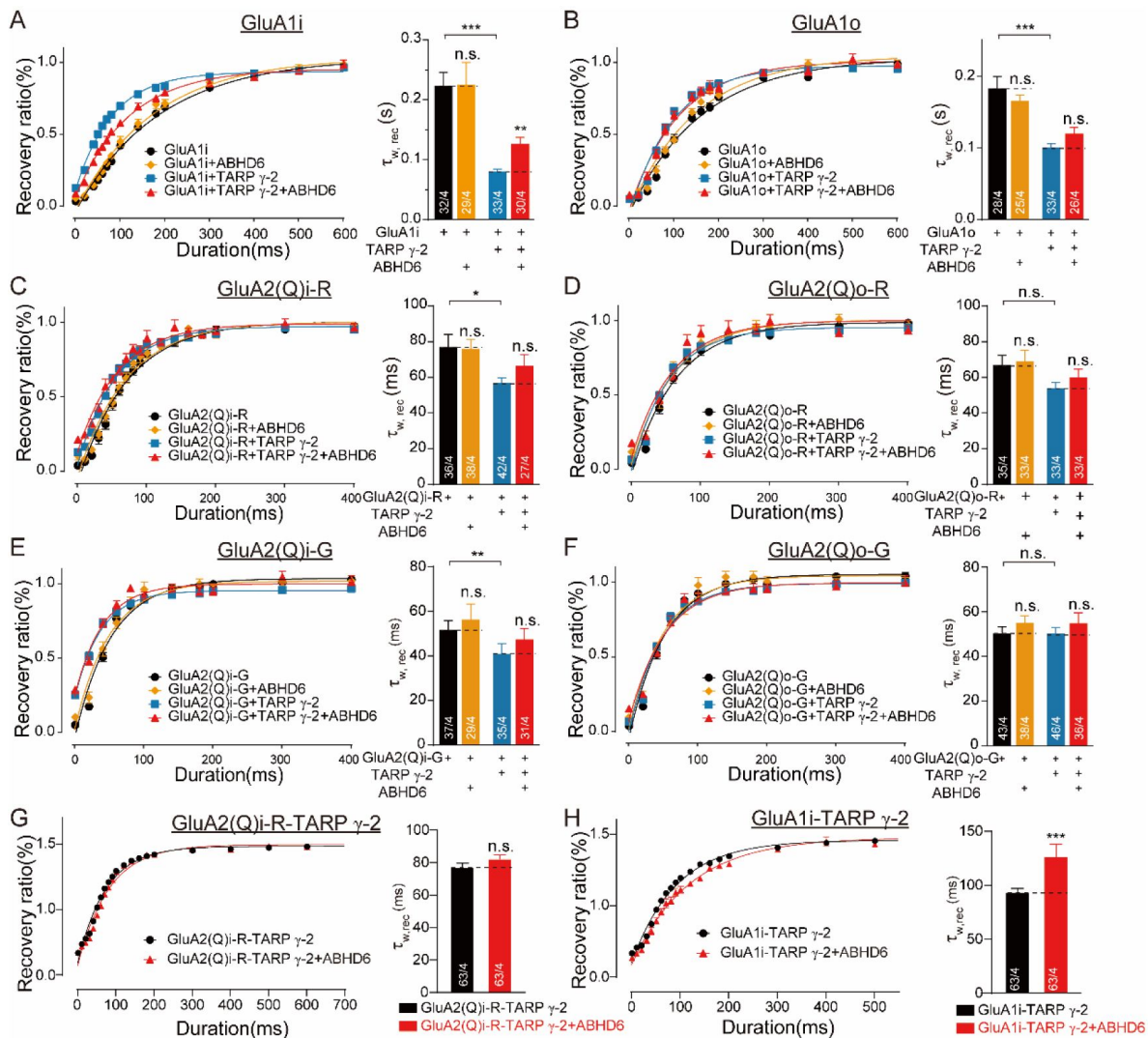


Figure 4.

Overexpression of ABHD6 slows the recovery from desensitization of GluA1i-TARP γ -2 complexes in HEK 293T cells.

(A-F) Glutamate (Glu, 10 mM) induced currents in an outside-out patch excised from an HEK 293T cell transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). The recovery ratio curves from desensitization, and the summary bar graphs of the $\tau_{w,rec}$. (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G) GluA2(Q)i-R-TARP γ -2 tandem. (H) GluA1i-TARP γ -2 tandem. The statistical method was one-way ANOVA followed by a two-way comparison (* P < 0.05; ** P < 0.01; *** P < 0.001. [Table EV4.2](#)).

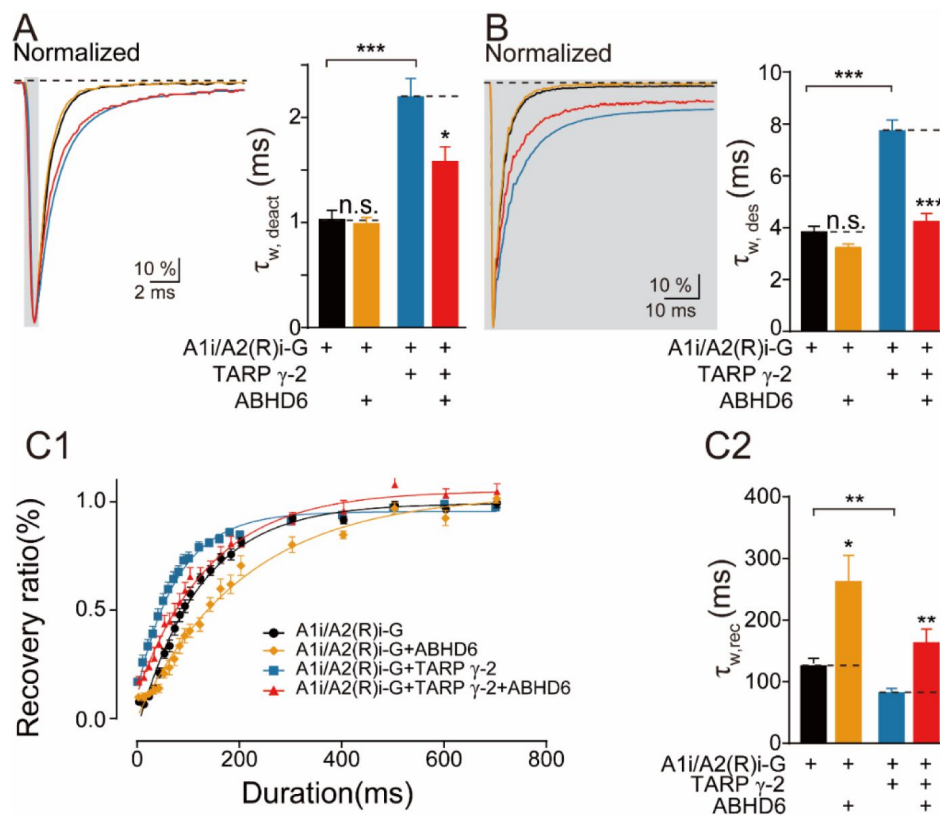


Figure 5.

Overexpression of ABHD6 accelerated the deactivation and desensitization of GluA1i/GluA2(R)i-G receptors-TARP γ -2 complexes in HEK 293T cells, slowed the recovery of GluA1i/GluA2(R)i-G receptors in the presence and absence of TARP γ -2.

(A) The normalized traces and the summary bar graphs of the $\tau_{w, deact}$ of Glutamate (10 mM Glu, 1 ms) induced currents in the outside-out patch recorded at -60 mV from HEK 293T cells transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red).

(B) The normalized traces and the summary bar graphs of the $\tau_{w, des}$ and peak amplitude of Glutamate (10 mM Glu, 100 ms) induced currents in the outside-out patch recorded at -60 mV from HEK 293T cells transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red).

(C) Glutamate (Glu, 10 mM) induced currents in an outside-out patch excised from an HEK 293T cell transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red). The first application of 100 ms glutamate was followed by a second glutamate application at increasing pulse intervals at -60 mV. The recovery ratio curves from desensitization (C1), and the summary bar graphs of the $\tau_{w, rec}$ (C2). The statistical method was one-way ANOVA followed by a two-way comparison (* $P < 0.05$; *** $P < 0.01$; **** $P < 0.001$. **Table. EV5.2** [↗](#))

ion-selective filter of the channel, causing impermeability to divalent cations (mainly Ca^{2+}), which not only affects the current and conductance of the channel but also plays an important role in neuronal functional impairment and excitotoxicity (Kawahara et al., 2004; Kwak and Kawahara, 2004). Our results showed that ABHD6 also inhibited currents in HEK 293T cells transfected with flop splice variants and R/G editing. Thus, this study provided further evidence that ABHD6 inhibited AMPARs on the amplitude of glutamate-induced currents. The reason for the decreased AMPAR-mediated currents might be due to the negative effects of ABHD6 on the tetramer formation of AMPARs reported by Schwenk J and colleagues (Schwenk et al., 2019).

AMPA receptors bind to multiple auxiliary subunits simultaneously, and the interactions between subunits directly influence the kinetic characteristics of AMPARs. TARPs γ -2, γ -3, γ -4, and γ -8 slow the deactivation and desensitization of GluA1-containing AMPARs in HEK 293T cells (Cho et al., 2007; Milstein et al., 2007). TARP γ -5 accelerates not only AMPAR deactivation and desensitization in HEK 293T cells but also desensitization after transfection into neurons (Kato et al., 2008). Studies have shown that in *Xenopus* oocytes and HEK cells, GSG1L slows GluA1/2i and GluA2(Q)i deactivation, desensitization, and recovery from desensitization and blocks the slowing effect of CNIH-2 on deactivation and desensitization (Schwenk et al., 2012; Shanks NF, 2012; Kamalova et al., 2020). GSG1L slows the kinetics of recombinant AMPARs in heterologous cells but accelerates those of natural AMPARs in CA1 pyramidal neurons when overexpressed in cultured organotypic hippocampal slices. Moreover, it slows the kinetics of CA1 pyramidal neurons in acute slices from knockout rats, which may be influenced by

AMPA subunit composition/chemometrics, posttranslational modifications, and auxiliary subunits in neurons superimposed on the co-expression of subunits (Gu et al., 2016; Mao et al., 2017). CNIH-2 and CNIH-3 slow GluA1i/ A2(R)i-G and GluA2(R)o-G/A4o-R deactivation and desensitization in *Xenopus* oocytes (Schwenk et al., 2009; Schwenk et al., 2012), as well as GluA1i, GluA2(Q)i, and GluA1i/A2(R)i deactivation and GluA1i desensitization in tsA201 and HEK 293T cells (Kato et al., 2010; Gill et al., 2011; Coombs et al., 2012). CNIH-3 overexpression slows AMPAR desensitization in cultured rat optic nerve oligodendrocyte precursor cells (Coombs et al., 2012), but CNIH-2 overexpression does not affect AMPAR kinetics in hippocampal pyramidal neurons and cerebellar granule neurons (Shi et al., 2010). Furthermore, CNIHs interact with TARPs. TARP γ -8 blocks the slowing effect of CNIH-2 on GluA2 but not GluA1 deactivation in HEK 293T cells (Herring et al., 2013). SynDIG4 can also slow GluA1i/GluA2(R)i deactivation and synergize with TARP γ -8, but does not affect GluA1/GluA2 heterodimer desensitization (Matt et al., 2018). CKAMP44 overexpression slows deactivation, accelerates desensitization, and slows recovery from desensitization, while CKAMP44 knockout has no effect on either deactivation or desensitization (Khodosevich et al., 2014; von Engelhardt, 2019). In this study, we systematically studied how ABHD6 affected the kinetic characteristics of AMPARs. In CA1 pyramidal neurons' synapses, the major form of AMPARs are heteromeric receptors that consisted of GluA2 with the other subunits (Malinow and Malenka, 2002; Brecht and Nicoll, 2003; Shepherd and Huganir, 2007). However, previous studies also found poorly expressed but significant population of GluA1 homomeric receptors in the hippocampus (Wenthold et al., 1996; Sans and C.Y., 2003). In addition, Zhao et al. studied the molecular structures of native AMPARs by cryo-EM, and elucidated the structure of 10 forms of native AMPARs which includes A2A2A2A2 homomeric form (Zhao et al., 2016). Thus, we studied the effect of ABHD6 on the kinetic characteristics of either homomeric AMPARs or heteromeric AMPARs. Our results clearly demonstrated that, unlike previously identified auxiliary subunits, ABHD6 accelerates GluA1 and GluA2(Q) deactivation and desensitization in the presence but not in the absence of TARP γ -2, which are also consistent with the previous publication that overexpression of ABHD6 accelerates AMPARs mediated deactivation in neurons (Wei et al., 2017). The results may suggest a novel mechanism by which auxiliary subunits regulate the kinetics of AMPARs and their ability not to regulate the kinetics of GluAs but to alter the kinetics of native AMPARs by altering the stoichiometry of TARPs in the AMPARs complex. Moreover, its effects do not depend on the

flip/flop isoform or R/G editing of AMPARs. We did not investigate the effects of ABHD6 on the kinetics of GluA2 Q607R-edited and GluA3-related AMPARs because their glutamate-induced currents are not detectable.

There are still questions that need to be answered. Whether ABHD6 also affects AMPAR kinetics in endogenously knocked out or overexpressed neurons? And what are the molecular mechanisms behind the role of ABHD6 in regulating AMPAR kinetics in a TARP γ -2-dependent manner? The structure of the AMPARs-TARP γ -2 complex has revealed that the TMD of TARP γ -2 is arranged around the TMD of AMPARs and serves to support the ion channel pore of the receptor (Chen et al., 2017 [↗](#); Zhao et al., 2019 [↗](#)). When AMPARs are activated, the differentially charged extracellular region of TARP γ -2 binds to the LBD of AMPARs, stabilizes the intra- and inter-dimeric interfaces of AMPARs, regulates the activation, deactivation and desensitization of AMPARs (Tomita et al., 2005 [↗](#); Zhao et al., 2016 [↗](#)). Previous studies have shown the binding of ABHD6 to the CTD of AMPARs through pull down experiments (Wei et al., 2017 [↗](#)). However, the CTD structure of TARP γ -2 and AMPARs has not been reported. Further structure-based studies may provide more information about the channel gating properties of AMPARs.

Materials and methods

Construction of expression vectors

Cloning of mouse ABHD6-2A-GFP containing pFUGW was performed as previously described. GluA1-3 cDNA (rat) applied to plasmid construction as previously described (Wei et al., 2017 [↗](#); Jiang et al., 2021 [↗](#)). The flip and flop isoforms were cloned into an IRES-GFP expression vector using polymerase chain reaction (PCR). Q/R and R/G editing variants were generated using PCR. Seventeen plasmids of GluA were constructed: GluA1i, GluA1o, GluA2(Q)i-R, GluA2(Q)o-R, GluA2(Q)i-G, GluA2(Q)o-G, GluA2(R)i-R, GluA2(R)o-R, GluA2(R)i-G, GluA2(R)o-G, GluA3i-R, GluA3o-R, GluA3i-G, GluA3o-G, GluA1i- γ -2, GluA2(Q)i- γ -2 and GluA1i/GluA2(R)i-G.

HEK 293T cell culture and transfection

HEK 293T cells were cultured in Dulbecco's Modified Eagle Medium (Gibco) containing 10% fetal bovine serum (ExCell Bio) and 1% penicillin–streptomycin (Gibco) in an incubator with 5% CO₂ at 37 °C. The transfection reagent used was polyethylenimine (Polysciences) (Chai et al., 2017 [↗](#); Wang et al., 2023 [↗](#); Wei et al., 2024 [↗](#)). The weight ratio of the GluA, TARP γ -2, and ABHD6 plasmids was 2:3:0.3125 (the GluA plasmid was 0.8 μ g/well). Transfection was terminated 3 h after initiation. HEK 293T cells were dissociated with 0.05% trypsin and then placed on poly-D-lysine-coated coverslips. To prevent cell toxicity due to AMPARs transfection, 200 μ M NBQX (Abcam) was added to the culture medium after transfection. The transfected cells were grown for > 18 h before electrophysiology recording.

Outside-out patch recording

Recordings were obtained using an EPC 10 patch clamp (HEKA). The data were digitized at 10 kHz. Unless otherwise stated, the holding potential was -60 mV, and series resistance compensation was set to 60–80%, and recordings with series resistances >20 M Ω or the I_{leak} > -200 pA were rejected. The cells were continuously perfused with an external solution containing 150 mM NaCl, 4 mM KCl, 2 mM CaCl₂, 1 mM MgCl₂, 10 mM HEPES, and 10 mM glucose at pH 7.4 and Osm 315. Electrodes with a resistance of 4–6 M Ω were pulled from borosilicate glass capillaries (WPI). The internal solution contained 125 mM KF, 33 mM KOH, 2 mM MgCl₂, 1 mM CaCl₂, 11 mM EGTA, and 10 mM HEPES at pH 7.4 and Osm 310, and 50 μ M spermine was added when indicated. Solutions containing L-glutamate (Sigma-Aldrich) were applied by patch membrane perfusion with θ tubes driven by a piezo manipulator (HVA). Glutamate (10 mM) pulses of 1 or 100 ms were applied every 5 s. To calculate the weighted time constant, time constant of deactivation ($\tau_{w, deact}$) and time

constant of desensitization ($\tau_{w,des}$) were fitted according to a double exponential function. For AMPAR recovery from desensitization, two-pulse (10 mM glutamate; 100 ms) stimulation was performed at intervals of 1–601 ms. To allow full recovery from desensitization, the sweeps were separated by 5 s. The current evoked by the preceding stimulus was defined as 1, while the other was defined as 2. The recovery ratio was calculated as follows:

$$Recovery\ ratio\ (\%) = (I_{Peak} - I_{steady-state})_2 / (I_{Peak} - I_{steady-state})_1$$

The time constant of recovery from desensitization (τ_{rec}) was calculated using nonlinear one-phase exponential decay fitting.

Statistical analysis

The data were analyzed using Clampfit 9.02 (Molecular Devices), Igor Pro 6.0 (WaveMetrics), and Prism 8 (GraphPad Software). Data normality was assessed using the D'Agostino–Pearson omnibus normality test. Statistical differences were examined using one-way ANOVA for normally distributed data or a nonparametric test otherwise. Correlation was examined using Pearson correlation analysis. Values of $P < 0.05$ were considered statistically significant.

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Author contributions

R.C., H.L., H.Y., J.G., L.Y., M.W., and C.Z. designed the research; R.C., H.L., H.Y., J.G., S.W. and M.W. conducted the research and analyzed the data; X.G., T.S., Y.Z., D.W. and X.C. analyzed the data; M.W. and C.Z. wrote the paper.

Competing interests

The authors declare that they have no conflicts of interest.

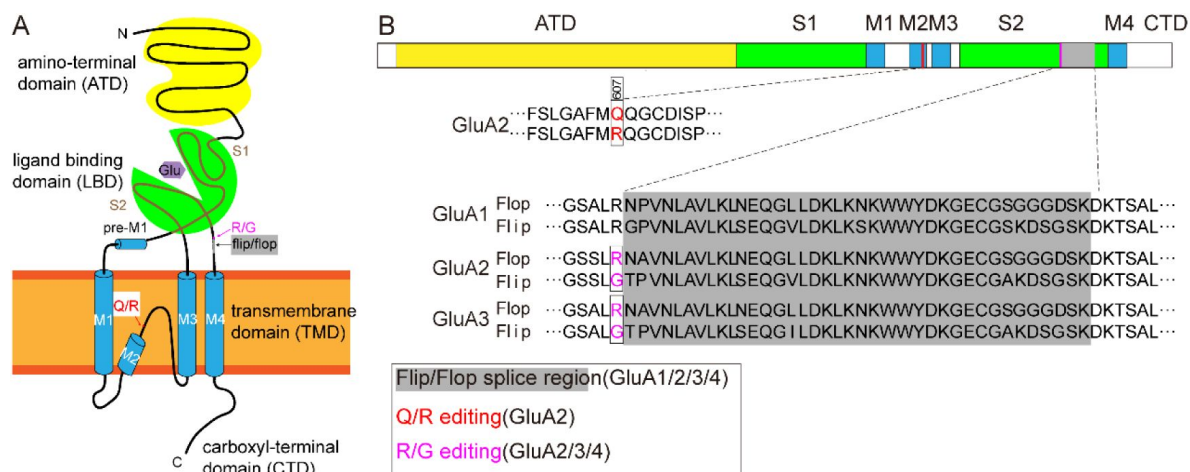


Figure EV1.

Schematic illustration of the AMPAR subunit and the Sequence alignment of RNA splice variants and editing of AMPAR.

(A) Topology of a single AMPAR subunit in the plasma membrane. Each subunit consists of an extracellular N-terminal domain (NTD), a ligand-binding domains (LBD), a transmembrane domain (TMD, M1–4) and an intracellular C-terminal domain (CTD), the flip/flop splice variants and the RNA editing sites (Q/R and R/G) are also shown.

(B) Sequence alignment of RNA splice variants and editing. Q/R editing sites (red letters) (GluA2), R/G editing sites (purple letters) (GluA2, A3), flip/flop splice variants (Grey box) (GluA1–A3). Complete sequences can be found in UniProt. Sequences are homologous and conserved in mouse, rat and human.

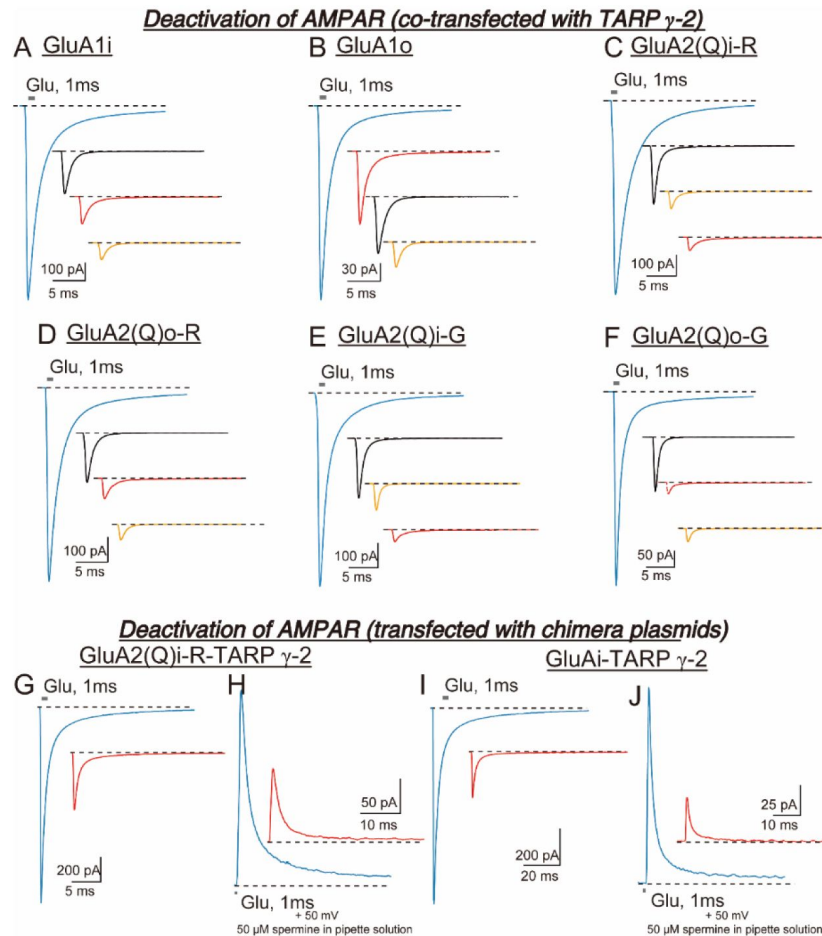


Figure EV2.

Average traces of deactivation of AMPAR with overexpression of ABHD6 in HEK 293T cells.

The average traces of Glutamate (10 mM Glu, 1 ms) induced currents in the outside-out patch from HEK 293T cells transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G, H) GluA2(Q)i-R-TARP γ -2 tandem. (I, J) GluA1i-TARP γ -2 tandem.

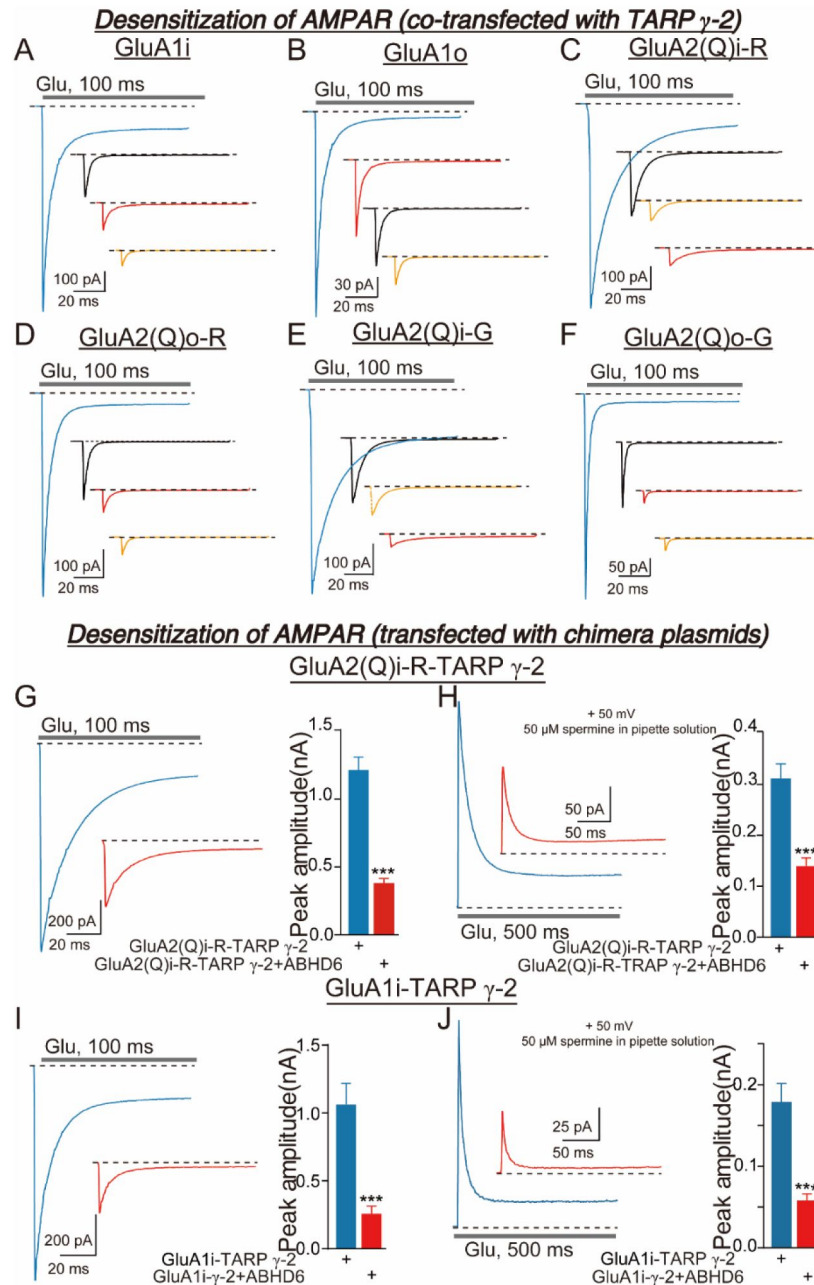


Figure EV3.

Average traces of desensitization of AMPAR with overexpression of ABHD6 in HEK 293T cells.

The average traces of Glutamate (10 mM Glu, 500 ms) induced currents in the outside-out patch from HEK 293T cells transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G, H) GluA2(Q)i-R-TARP γ -2 tandem. (I, J) GluA1i-TARP γ -2 tandem.

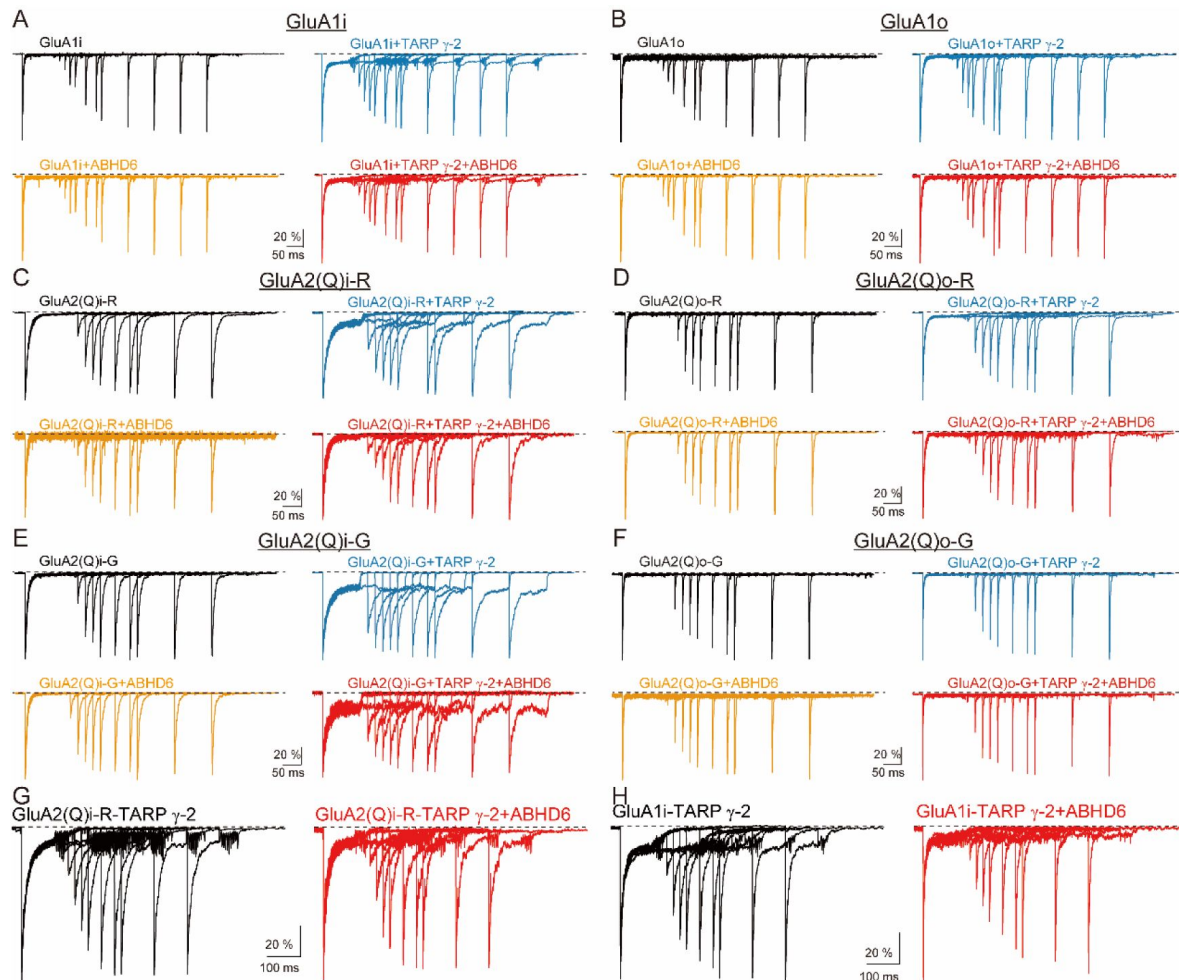


Figure EV4.

Typical traces of the recovery from desensitization of AMPAR in HEK 293T cells.

(A-F) Glutamate (Glu, 10 mM) induced currents in an outside-out patch excised from an HEK 293T cell transfected with GluA (black), GluA + ABHD6 (orange), GluA + TARP γ -2 (blue), and GluA + TARP γ -2 + ABHD6 (red). The first application of 100 ms glutamate was followed by a second glutamate application at increasing pulse intervals at -60 mV. The typical traces from a cell are normalized and aligned to the peak. The typical traces of the recovery from desensitization. (A) GluA1i. (B) GluA1o. (C) GluA2(Q)i-R. (D) GluA2(Q)o-R. (E) GluA2(Q)i-G. (F) GluA2(Q)o-G. (G) GluA2(Q)i-R-TARP γ -2 tandem. (H) GluA1i-TARP γ -2 tandem.

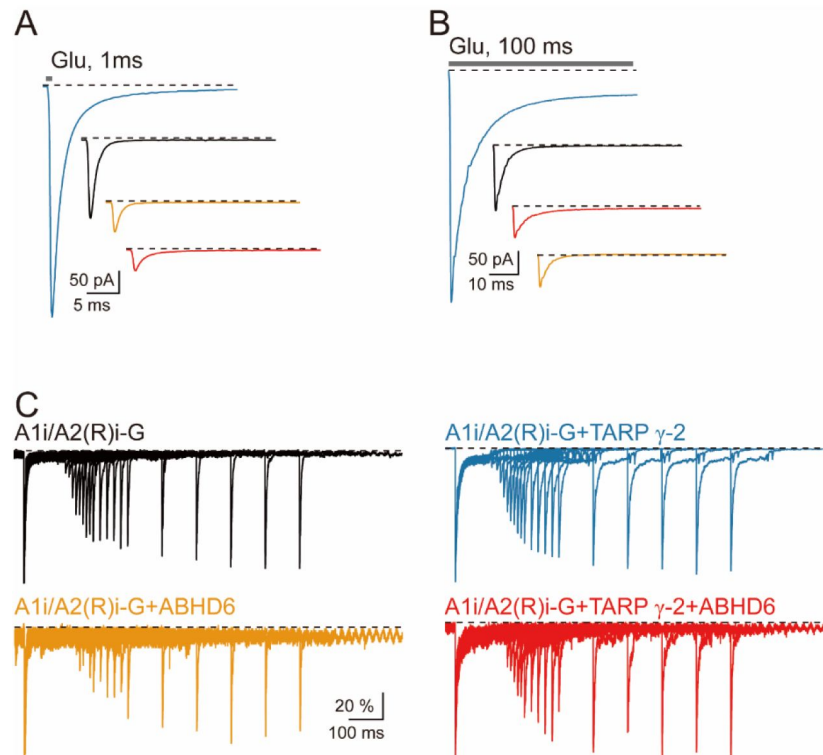


Figure EV5.

Average traces of the deactivation, desensitization and recovery from desensitization of GluA1i/GluA2(R)i-G receptors-TARP γ -2 complexes in HEK 293T cells.

(A) The average traces of the $\tau_{w, deact}$ of Glutamate (10 mM Glu, 1 ms) induced currents in the outside-out patch recorded at -60 mV from HEK 293T cells transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red).

(B) The average traces of the $\tau_{w, des}$ and peak amplitude of Glutamate (10 mM Glu, 100 ms) induced currents in the outside-out patch recorded at -60 mV from HEK 293T cells transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red).

(C) The typical trace of recovery from desensitization. Glutamate (Glu, 10 mM) induced currents in an outside-out patch excised from an HEK 293T cell transfected with GluA1i/GluA2(R)i-G (black), GluA1i/GluA2(R)i-G + ABHD6 (orange), GluA1i/GluA2(R)i-G + TARP γ -2 (blue), and GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6 (red). The first application of 100 ms glutamate was followed by a second glutamate application at increasing pulse intervals at -60 mV.

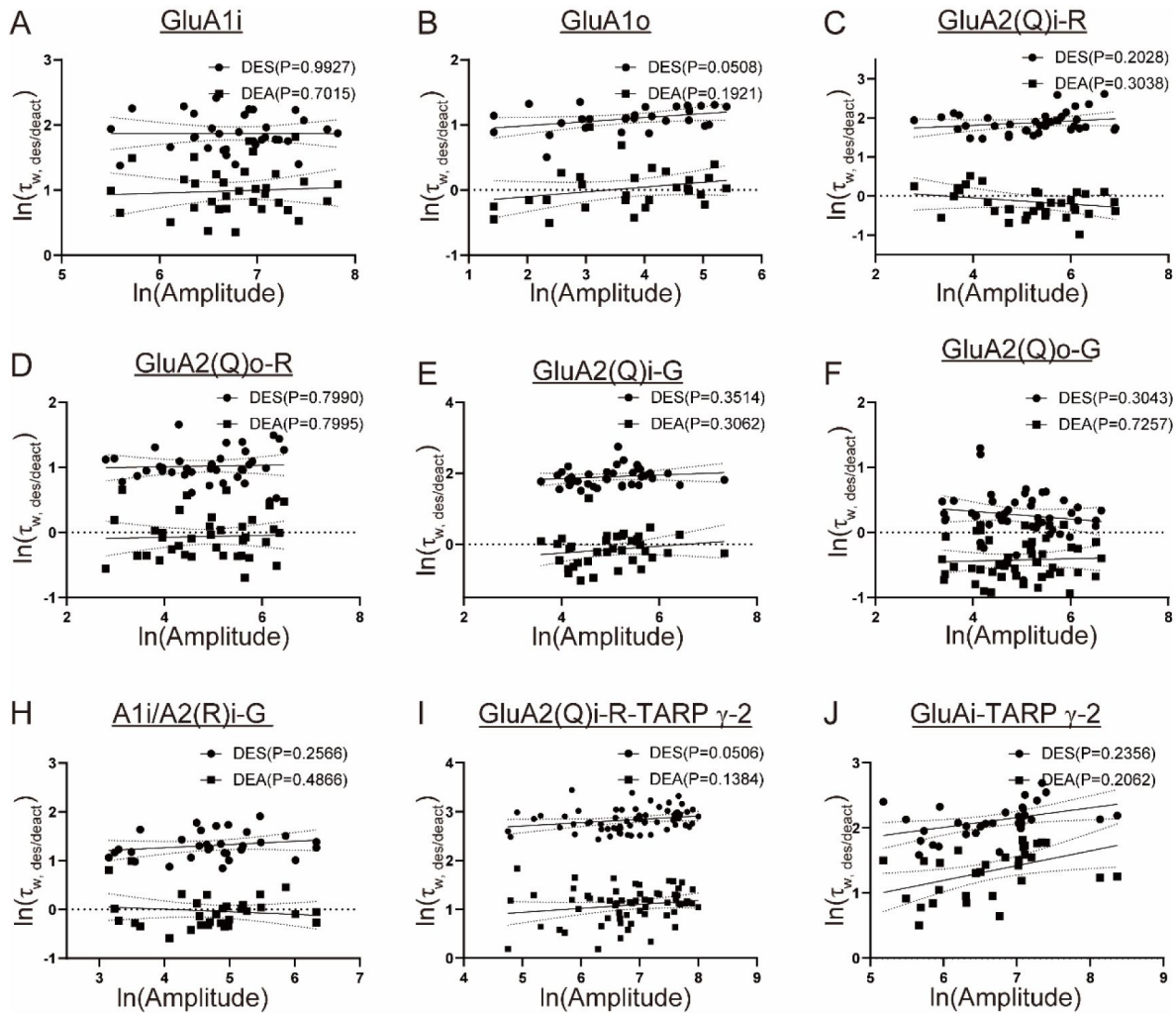


Figure EV6.

Pearson's correlation between natural logarithm peak amplitude (pA) and the $\tau_{w,deact}$ and $\tau_{w,des}$ coexpression with various GluA subunits.

(A-J) Pearson's correlation and Local Polynomial Regression (loess) with 95% confidence intervals between natural logarithm peak amplitude (pA) and the $\tau_{w,deact}$ (ms) of Glutamate (10 mM Glu, 1 ms) induced currents and $\tau_{w,des}$ (ms) of Glutamate (10 mM Glu, 100 ms) induced currents in the outside-out patch from HEK 293T cells transfected with various GluA subunits.

Subnit	GluA		GluA + ABHD6		GluA + TARP γ -2		GluA + TARP γ -2 + ABHD6	
GluA1i	226.69 167.55(32)	$\pm$	77.91 76.58(36) ^a	$\pm$	1005.89 530.81(33) ^a	$\pm$	135.92 140.83(36) ^b	$\pm$
GluA1o	88.67 59.73(33)	$\pm$	42.62 48.11(38) ^a	$\pm$	309.24 211.47(37) ^a	$\pm$	115.51 126.12(30) ^b	$\pm$
GluA2(Q)i-R	281.51 256.66(36)	$\pm$	87.50 81.75(42) ^a	$\pm$	911.62 607.39(38) ^a	$\pm$	77.86 157.14(44) ^b	$\pm$
GluA2(Q)o-R	201.23 171.09(38)	$\pm$	63.84 95.32(46) ^a	$\pm$	736.02 579.63(34) ^a	$\pm$	82.15 205.29(41) ^b	$\pm$
GluA2(Q)i-G	211.76 254.32(38)	$\pm$	95.45 119.83(38) ^a	$\pm$	660.77 389.82(38) ^a	$\pm$	42.89 59.48(43) ^b	$\pm$
GluA2(Q)o-G	196.15 175.99(48)	$\pm$	41.57 39.21(45) ^a	$\pm$	567.72 361.19(46) ^a	$\pm$	37.75 56.79(49) ^b	$\pm$
GluA2(R)i-R	2.55 $\pm$ 3.32(26)		1.86 1.27(30)	$\pm$	55.83 52.57(39) ^a	$\pm$	5.34 6.69(30) ^b	$\pm$
GluA2(R)o-R	1.72 $\pm$ 1.51(14)		1.72 0.88(29)	$\pm$	28.69 32.66(34) ^a	$\pm$	12.63 19.49(26) ^b	$\pm$
GluA2(R)i-G	2.60 $\pm$ 3.59(33)		2.55 2.56(33)	$\pm$	31.32 35.76(36) ^a	$\pm$	3.29 4.82(33) ^b	$\pm$
GluA2(R)o-G	2.57 $\pm$ 2.41(43)		1.95 1.62(49)	$\pm$	9.26 9.75(48) ^a	$\pm$	1.95 1.52(49) ^b	$\pm$
GluA3i-R	21.11 40.80(26)	$\pm$	3.29 5.57(20)	$\pm$	24.28 46.24(25)	$\pm$	1.22 1.15(21) ^b	$\pm$
GluA3o-R	6.10 $\pm$ 5.89(25)		2.46 2.27(20)	$\pm$	11.38 15.56(25)	$\pm$	4.36 11.04(21) ^b	$\pm$
GluA3i-G	9.83 16.73(25)	$\pm$	3.32 4.01(19)	$\pm$	15.27 17.49(26) ^a	$\pm$	3.25 2.83(19) ^b	$\pm$
GluA3o-G	3.12 $\pm$ 4.66(25)		1.15 0.67(20)	$\pm$	4.70 7.75(26)	$\pm$	1.15 0.95(20) ^b	$\pm$
GluA2(Q)i-R-TARP γ -2	3.23 1.15(73)		$\pm$ 2.34 (73)	$\pm$ 1.27	\		\	
GluA1i-TARP γ -2	4.19 1.69(36)		$\pm$ 2.31 (36)	$\pm$ 1.11	\		\	
GluA2(Q)i-R-TARP γ -2 (+ 50 mV)	2.81 1.80(74)		$\pm$ 1.71 0.93(71)	$\pm$	\		\	
GluA1i-TARP γ -2 (+ 50 mV)	2.64 1.23(36)		$\pm$ 1.86 1.05(36)	$\pm$	\		\	

Mean ($\pm$ S.D.) values for the peak amplitude of glutamate-induced currents which were calculated from the desensitization curves. And the curves from a double exponential function fits to the glutamate-induced currents evoked by 100 ms applications of 10 mM glutamate. Currents were recorded in outside-out patch

Table EV1.1.

Summary of peak amplitude(pA) of GluAs when co-transfected with/without γ -2 or/and ABHD6.

recordings (-60 mV) from cells transfected with GluA, GluA + ABHD6, GluA + TARP γ -2, GluA + TARP γ -2 + ABHD6. The number of patches is given in parenthesis.

^aSignificantly different from GluA.

^bSignificantly different from GluA + TARP γ -2.

Table EV1.1. (continued)

Subnit	GluA+ABHD6 vs. GluA	GluA+TARP vs. GluA	γ -2 GluA+TARP 2+ABHD6 vs. GluA+TARP γ -2
GluA1i	P=0.0037(**)	P<0.0001(***)	P<0.0001(***)
GluA1o	P=0.0143(*)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)i-R	P=0.0045(**)	P=0.0009(***)	P<0.0001(***)
GluA2(Q)o-R	P=0.0001(***)	P=0.0018(**)	P<0.0001(***)
GluA2(Q)i-G	P=0.0070(**)	P=0.0003(***)	P<0.0001(***)
GluA2(Q)o-G	P<0.0001(***)	P=0.0008(***)	P<0.0001(***)
GluA2(R)i-R	P>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(R)o-R	P>0.9999(-)	P<0.0001(***)	P=0.0485(*)
GluA2(R)i-G	P>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(R)o-G	P=0.1356(-)	P<0.0001(***)	P<0.0001(***)
GluA3i-R	P=0.0007(-)	P>0.9999(-)	P<0.0001(***)
GluA3o-R	P=0.0775(-)	P>0.9999(-)	P=0.0028(**)
GluA3i-G	P>0.9999(-)	P=0.0066(**)	P=0.0005(***)
GluA3o-G	P>0.9999(-)	P=0.0838(-)	P=0.0002(***)
GluA2(Q)i-R- TARP γ -2	P<0.001(***)	\	\
GluA1i-TARP γ -2	P<0.001(***)	\	\
GluA2(Q)i-R- TARP γ -2 (+ 50 mV)	P<0.001(***)	\	\
GluA1i-TARP γ -2 (+ 50 mV)	P<0.001(***)	\	\

P values for comparison of peak amplitude of GluAs when co-transfected with/without γ -2 or/and ABHD6. The data were analyzed using Prism 8 (GraphPad Software), and each p values was adjusted to account for multiple comparisons.

Table EV1.2.

Summary of P values for comparison of peak amplitude.

Subnit	GluA	GluA + ABHD6	GluA + TARP γ -2	GluA + TARP γ -2 + ABHD6
GluA1i	0.99 $\pm$ 0.21(32)	0.98 0.41(36)	$\pm$ 2.88 1.16(33) ^a	$\pm$ 1.38 $\pm$ 0.38(36) ^b
GluA1o	1.09 $\pm$ 0.27(33)	1.00 0.35(36)	$\pm$ 1.91 0.81(37) ^a	$\pm$ 1.25 0.49(30) ^b
GluA2(Q)i-R	1.03 $\pm$ 1.05(36)	1.15 $\pm$ 0.77(43)	3.68 3.13(38) ^a	$\pm$ 1.78 1.39(43) ^b
GluA2(Q)o-R	0.97 $\pm$ 0.40(38)	1.07 0.42(47)	$\pm$ 2.78 1.43(34) ^a	$\pm$ 1.66 0.95(46) ^b
GluA2(Q)i-G	0.96 $\pm$ 0.55(37)	0.81 0.37(38)	$\pm$ 2.44 1.46(38) ^a	$\pm$ 1.79 1.26(44)
GluA2(Q)o-G	0.68 $\pm$ 0.21(48)	0.76 0.27(47)	$\pm$ 1.90 1.30(46) ^a	$\pm$ 1.14 0.60(50) ^b
GluA2(Q)i-R-TARP γ -2	17.47 $\pm$ 4.68(73)	11.78 $\pm$ 3.20(73)		\
GluA1i-TARP γ -2	8.78 $\pm$ 2.76(36)	5.80 $\pm$ 2.12(36)		\
GluA2(Q)i-R-TARP γ -2 (+ 50 mV)	15.52 $\pm$ 4.48(74)	10.31 3.71(71)	$\pm$	\
GluA1i-TARP γ -2 (+ 50 mV)	6.81 $\pm$ 2.01(36)	5.64 1.85(36)	$\pm$	\

Mean ($\pm$ S.D.) values for the time constant of deactivation ($\tau_{w, deact}$). The deactivation curves from a double exponential function fits to the glutamate-induced currents evoked by 1 ms applications of 10 mM glutamate. Currents were recorded in outside-out patch recordings (-60 mV) from cells transfected with GluA, GluA + ABHD6, GluA + TARP γ -2, GluA + TARP γ -2 + ABHD6. The number of patches is given in parenthesis.

^aSignificantly different from GluA.

^bSignificantly different from GluA + TARP γ -2.

Table EV2.1.

Summary of $\tau_{w, deact}$ (ms) of GluAs when co-transfected with/without γ -2 or/and ABHD6.

Subnit	GluA+ABHD6 vs. GluA	GluA+TARP vs. GluA	γ -2	GluA+TARP 2+ABHD6 vs. GluA+TARP γ -2
GluA1i	P>0.9999(-)	P<0.0001(***)		P<0.0001(***)
GluA1o	P>0.9999(-)	P<0.0001(***)		P=0.0005(***)
GluA2(Q)i-R	P>0.9999(-)	P<0.0001(***)		P=0.0003(***)
GluA2(Q)o-R	P>0.9999(-)	P<0.0001(***)		P=0.0012(**)
GluA2(Q)i-G	P>0.9999(-)	P<0.0001(***)		P=0.1473(-)
GluA2(Q)o-G	P>0.9999(-)	P<0.0001(***)		P=0.0005(***)
GluA2(Q)i-R- TARP γ -2	P<0.001(***)	\		\
GluA1i-TARP γ - 2	P<0.001(***)	\		\
GluA2(Q)i-R- TARP γ -2 (+ 50 mV)	P<0.001(***)	\		\
GluA1i-TARP γ - 2 (+ 50 mV)	P<0.001(***)	\		\

P values for comparison of $\tau_{w, deact}$ of GluAs when co-transfected with/without γ -2 or/and ABHD6. The data were analyzed using Prism 8 (GraphPad Software), and each p values was adjusted to account for multiple comparisons.

Table EV2.2.

Summary of P values for comparison of $\tau_{w, deact}$.

Subnit	GluA	GluA + ABHD6	GluA + TARP γ -2	GluA + TARP γ -2 + ABHD6
GluA1i	2.91 $\pm$ 0.53(32)	2.59 0.78(36)	$\pm$ 6.72 1.86(33) ^a	$\pm$ 4.14 1.35(36) ^b
GluA1o	3.39 $\pm$ 0.70(33)	2.90 0.69(38)	$\pm$ 5.52 1.59(37) ^a	$\pm$ 3.50 1.15(30) ^b
GluA2(Q)i-R	6.82 $\pm$ 2.19(36)	6.26 1.83(42)	$\pm$ 15.43 4.87(38) ^a	$\pm$ 9.13 3.41(44) ^b
GluA2(Q)o-R	2.82 $\pm$ 0.81(38)	2.66 0.78(46)	$\pm$ 4.71 1.48(34) ^a	$\pm$ 3.08 1.33(41) ^b
GluA2(Q)i-G	7.00 $\pm$ 2.11(38)	6.25 $\pm$ 2.0(38)	15.85 3.95(38) ^a	$\pm$ 9.52 3.93(43) ^b
GluA2(Q)o-G	1.36 $\pm$ 0.58(48)	1.36 0.51(45)	$\pm$ 2.01 0.60(46) ^a	$\pm$ 1.43 0.50(49) ^b
GluA2(Q)i-R-TARP γ -2	17.47 $\pm$ 4.68(73)	11.78 $\pm$ 3.20(73)	\	\
GluA1i-TARP γ -2	8.78 $\pm$ 2.76(36)	5.80 $\pm$ 2.12(36)	\	\
GluA2(Q)i-R-TARP γ -2 (+ 50 mV)	15.52 $\pm$ 4.48(74)	10.31 3.71(71)	$\pm$ \	\
GluA1i-TARP γ -2 (+ 50 mV)	6.81 $\pm$ 2.01(36)	5.64 1.85(36)	$\pm$ \	\

Mean ($\pm$ S.D.) values for time constant of desensitization ($\tau_{w, des}$) from the desensitization curves. Currents were recorded in outside-out patch recordings (-60 mV) from cells transfected with GluA, GluA + ABHD6, GluA + TARP- γ -2, GluA + TARP- γ -2 + ABHD6. The number of patches is given in parenthesis.

^aSignificantly different from GluA.

^bSignificantly different from GluA + TARP γ -2.

Table EV3.1.

Summary of $\tau_{w, des}$ (ms) of GluAs when co-transfected with/without γ -2 or/and ABHD6.

Subnit	GluA+ABHD6 vs. GluA	GluA+TARP vs. GluA	γ -2 GluA+TARP 2+ABHD6 vs. GluA+TARP γ -2
GluA1i	P>0.9999(-)	P<0.0001(***)	P=0.0004(***)
GluA1o	P=0.2467(-)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)i-R	P>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)o-R	P>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)i-G	P>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)o-G	P=>0.9999(-)	P<0.0001(***)	P<0.0001(***)
GluA2(Q)i-R- TARP γ -2	P<0.001(***)	\	\
GluA1i-TARP γ - 2	P<0.001(***)	\	\
GluA2(Q)i-R- TARP γ -2 (+ 50 mV)	P<0.001(***)	\	\
GluA1i-TARP γ - 2 (+ 50 mV)	P=0.013(*)	\	\

P values for comparison of $\tau_{w, des}$ of GluAs when co-transfected with/without γ -2 or/and ABHD6. The data were analyzed using Prism 8 (GraphPad Software), and each p values was adjusted to account for multiple comparisons.

Table EV3.2.

Summary of P values for comparison of $\tau_{w, des}$

Subnit	GluA		GluA + ABHD6		GluA + TARP γ -2		GluA + TARP γ -2 + ABHD6	
GluA1i	223.90	±	225.00	±	80.66	±	126.50	±
	124.60(32)		201.39(29)		22.56(33) ^a		58.83(30) ^b	
GluA1o	182.60	±	165.60	±	100.70	±	119.90	±
	87.20(28)		39.86(25)		27.76(33) ^a		41.56(26)	
GluA2(Q)i-R	77.17	±	75.89	±	56.98	±	66.44	±
	41.26(36)		32.54(38)		17.86(42) ^a		33.63(27)	
GluA2(Q)o-R	67.60	±	68.87	±	54.05	±	59.95	±
	28.29(35)		36.06(33)		17.56(33)		26.61(33)	
GluA2(Q)i-G	52.29	±	56.23	±	40.85	±	47.39	±
	20.93(37)		37.72(29)		27.86(35) ^a		26.72(31)	
GluA2(Q)o-G	50.48	±	58.12	±	50.39	±	55.68	±
	16.68(44)		33.49(39)		18.40(45)		29.88(35)	
GluA2(Q)i-R-TARP γ -2	77.61± 20.99(63)		82.66± 20.75(63)		\		\	
GluA1i-TARP γ -2	93.89± 22.90(36)		127.22± 11.96(36)		\		\	

Mean ($\pm$ S.D.) values for constant of recovery from desensitization ($\tau_{w, rec}$) from the recovery ratio curves, which were calculated using nonlinear one-phase exponential decay fitting. Currents were recorded in outside-out patch recordings (-60 mV, 10 mM glutamate; 100 ms) when performing two-pulse stimulation, from cells transfected with GluA, GluA + ABHD6, GluA + TARP γ -2, GluA + TARP γ -2 + ABHD6. The number of patches is given in parenthesis.

^aSignificantly different from GluA.

^bSignificantly different from GluA + TARP γ -2.

Table EV4.1.

Summary of $\tau_{w, rec}$ (ms) of GluAs when co-transfected with/without γ -2 or/and ABHD6.

Subnit	GluA+ABHD6 vs. GluA	GluA+TARP vs. GluA	γ -2	GluA+TARP 2+ABHD6 vs. GluA+TARP γ -2
GluA1i	P>0.9999(-)	P<0.0001(***)		P=0.0092(**)
GluA1o	P>0.9999(-)	P<0.0001(***)		P=0.4938(-)
GluA2(Q)i-R	P>0.9999(-)	P=0.0325(*)		P>0.9999(-)
GluA2(Q)o-R	P>0.9999(-)	P=0.1988(-)		P>0.9999(-)
GluA2(Q)i-G	P>0.9999(-)	P=0.0014(**)		P=0.8413(-)
GluA2(Q)o-G	P>0.9999(-)	P>0.9999(-)		P>0.9999(-)
GluA2(Q)i-R- TARP γ -2	P=0.075(-)	\		\
GluA1i-TARP γ - 2	P<0.001(***)	\		\

P values for comparison of $\tau_{w, rec}$ of GluAs when co-transfected with/without γ -2 or/and ABHD6. The data were analyzed using Prism 8 (GraphPad Software), and each p values was adjusted to account for multiple comparisons.

Table EV4.2.

Summary of P values for comparison of $\tau_{w, rec}$

	GluA1i/GluA2(R)i-G		GluA1i/GluA2(R)i-G + ABHD6		GluA1i/GluA2(R)i-G + TARP γ -2		GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6	
Peak amplitude	156.96	$\pm$	65.75	$\pm$	492.72	$\pm$	67.45	$\pm$
de	140.07(31)		66.79(33) ^a		454.74(34) ^a		141.71(32) ^b	
$\tau_{w, deact}$	1.04 $\pm$ 0.46(31)		1.00 $\pm$ 0.30(33)		2.20 $\pm$ 0.98(34) ^a		1.59 $\pm$ 0.76(32) ^b	
$\tau_{w, des}$	3.85 $\pm$ 1.10(31)		3.24 $\pm$ 0.76(33)		7.77 $\pm$ 2.27(34) ^a		4.27 $\pm$ 1.53(32) ^b	
τ_{rec}	129.72	$\pm$	263.71	$\pm$	83.11	$\pm$	164.42	$\pm$
	39.79(23)		180.45(19) ^a		28.70(23) ^a		86.02(17) ^b	

Mean ($\pm$ S.D.) values for peak amplitude, $\tau_{w, des}$ from the desensitization and $\tau_{w, deact}$ from the deactivation curves, and τ_{rec} from the recovery ratio curves. Currents were recorded in outside-out patch recordings (-60 mV, 10 mM glutamate; 100 ms / 1 ms / 100 two-pulse stimulation) without spermine in internal solution, from cells transfected with GluA1i/GluA2(R)i-G, GluA1i/GluA2(R)i-G + ABHD6, GluA1i/GluA2(R)i-G + TARP γ -2, GluA1i/GluA2(R)i-G + TARP γ -2 + ABHD6. The number of patches is given in parenthesis.

^aSignificantly different from GluA1i/GluA2(R)i-G.

^bSignificantly different from GluA1i/GluA2(R)i-G + TARP γ -2.

Table EV5.1.

Summary of GluA1i/GluA2(R)i-G receptors when co-transfected with/without γ -2 or/and ABHD6.

	GluA1i/GluA2(R)i-G +ABHD6 vs. GluA1i/GluA2(R)i-G	GluA1i/GluA2(R)i-G +TARP γ-2 vs. GluA1i/GluA2(R)i-G	GluA1i/GluA2(R)i-G +TARP γ-2+ABHD6 vs. GluA1i/GluA2(R)i-G +TARP γ-2
Peak amplitude	P=0.01(*)	Pp=0.007(**)	P<0.001 (***)
$\tau_{w, deact}$	P>0.99(-)	P<0.001(***)	P=0.03(*)
$\tau_{w, des}$	P=0.43(-)	P<0.001(***)	P<0.001(***)
τ_{rec}	P=0.0109(*)	P=0.0076(**)	P=0.0011(**)

P values for comparison of Peak amplitude, $\tau_{w, deact}$, $\tau_{w, des}$, τ_{rec} of GluA1i/GluA2(R)i-G when co-transfected with/without γ -2 or/and ABHD6. The data were analyzed using Prism 8 (GraphPad Software), and each p values was adjusted to account for multiple comparisons.

Table EV5.2.

Summary of P values for comparison of GluA1i/GluA2(R)i-G receptors when co transfected with/without γ -2 or/and ABHD6.

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Editors

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Lu Chen

Stanford University, Stanford, United States of America

Reviewer #1 (Public Review):

Summary:

This research sheds light on the nuanced role of ABHD6 in the regulation of AMPARs, highlighting its interaction with TARP γ -2 as a critical factor in modulating receptor-gating kinetics. It is crucial to understand that while ABHD6 alone does not alter AMPAR kinetics, its presence alongside TARP γ -2 leads to accelerated deactivation and desensitization of AMPARs, impacting synaptic transmission dynamics.

Strengths:

Important findings in the research include:

- ABHD6 does not affect the gating kinetics of GluA1 and GluA2(Q) homomeric receptors independently.
- In the presence of TARP γ -2, ABHD6 accelerates deactivation and desensitization of these receptors, regardless of their splicing or editing isoforms.
- The effect is consistent for both homomeric GluA1 and GluA2(Q) receptors and heteromeric GluA1i/GluA2(R)i-G receptors.
- The recovery from desensitization of GluA1 with the flip splicing isoform is slowed by ABHD6 in the presence of TARP γ -2.

Weaknesses:

However, the study focuses on specific receptor subunits and isoforms, which may not fully represent the diversity of AMPAR compositions found in vivo (e.g. though the authors have claimed that TARP γ -2 failed to increase GluA3-induced currents significantly, the effect on GluA4 or the explanation was missing). Further research is needed to explore the implications of these findings in more complex neuronal environments.

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Reviewer #2 (Public Review):

Summary:

Cong et al. investigated the regulatory effects of ABHD6 on AMPARs. The authors performed adequate electrophysiology recordings to show the exact pattern of this regulation and covered major critical points.

Strengths:

The authors have performed high-quality ephys recordings and examined all potential regulatory aspects of ABHD6 on AMPARs. This is important to understand the AMPAR functions.

Weaknesses:

- (1) The authors discussed CNIH-2 extensively from line 92-110 in the introduction, however, they did not perform related experiments. I suggest they move this part to the discussion where they also discussed the roles of CNIH.
- (2) The authors need to report the "n" for all the experiments they have presented in this manuscript. How many cells were recorded in each condition? How many batches? This

information has to be in all of the figure legends, but it is missing except Fig. 4.

(3) One question is what the physiological meanings of this regulatory effect are. The authors may consider adding some discussions.

(4) About statistics. The authors need to add more details and make sure their statistics sound. For example, they also need to check the equality of variances. In their Table EVs, where the P values are reported, the authors need to report which statistics they have used, one-way ANOVA, K-W test, or others, and the exact post-hoc test type for each comparison. For one-way ANOVA, report the F values simultaneously with the P values in all figure legends.

(5) Fig. 3J, the authors need to correct the label of the Y axis. It is shifted.

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